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(54) **ELECTRONIC DEVICE, CONTROL METHOD OF ELECTRONIC DEVICE, AND NON-TRANSITORY COMPUTER READABLE MEDIUM**

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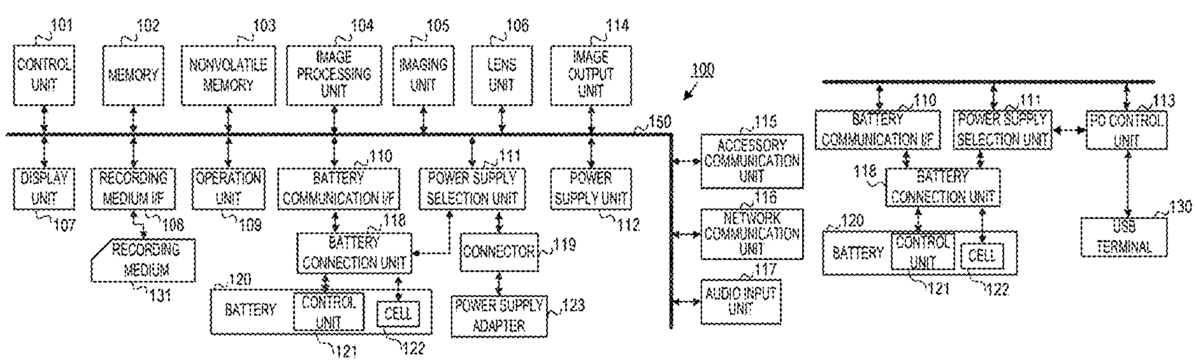
(57) **ABSTRACT**

In a case where a first power supply and a second power supply are connected to an electronic device, the electronic device acquires first information on the first power supply and second information on the second power supply, and determines, among the first power supply and the second power supply, a power supply having a larger suppliable electric power as a power supply to be used for an operation of the electronic device, and in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device, the electronic device sets a first operation mode if a first electric power that is suppliable by the first power supply is larger than a threshold power, and sets a second operation mode if the first electric power is smaller than the threshold power.



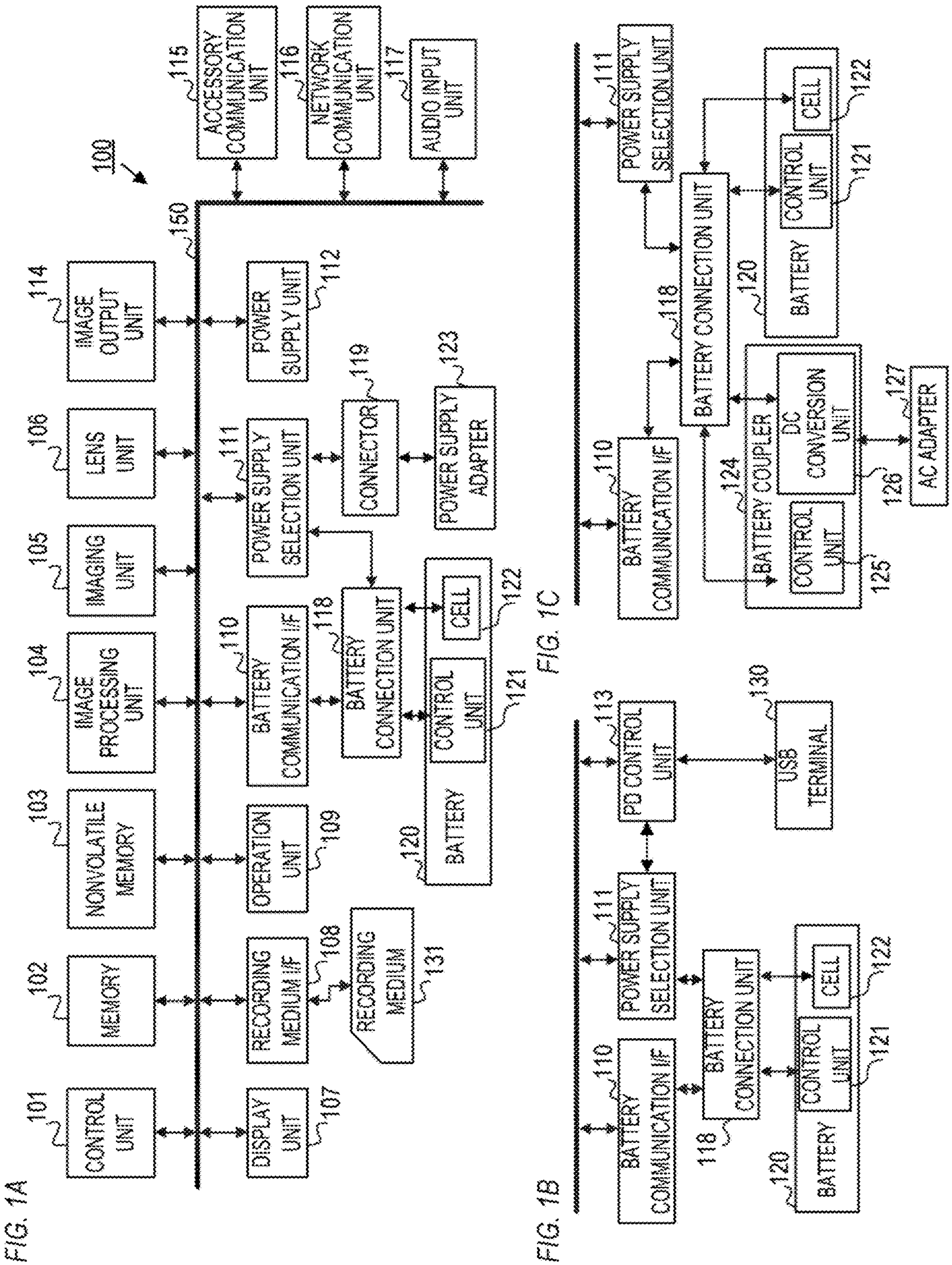


FIG. 1A
FIG. 1B
FIG. 1C

FIG. 2

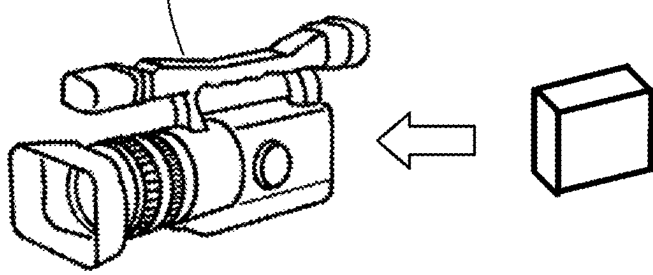
201

FUNCTION	POWER CONSUMPTION [W]	
	TURNED ON	TURNED OFF
NETWORK	5	0.3
ACCESSORY SHOE POWER SUPPLY	3	0
EXTERNAL MICROPHONE INPUT	2	0
LENS POWER SUPPLY	8	0.3
...	...	...
OTHERS	50	/

202

OPERATION MODE	NETWORK	ACCESSORY SHOE POWER SUPPLY	EXTERNAL MICROPHONE INPUT	LENS POWER SUPPLY	...	TOTAL ELECTRIC POWER
OPERATION MODE1	ON	ON	ON	ON	...	P1
OPERATION MODE2	ON	OFF	OFF	ON	...	P2
OPERATION MODE3	ON	OFF	ON	ON	...	P3
...	...	...	...	...	...	...

FIG. 3



301

INFORMATION ACQUIRED FROM BATTERY
BATTERY ID
CAPACITY [%]
VOLTAGE [V]
CONSUMPTION CURRENT [mA]
DISCHARGEABLE REMAINING AMOUNT [mAh]

302

ID	RATED POWER [W]	RATED VOLTAGE [V]	RATED CURRENT [A]
0×0001	90	12.0	8.0
0×0002	50	12.0	4.8
...	...	...	...

311

INFORMATION ACQUIRED FROM BATTERY COUPLER
BATTERY ID
VOLTAGE [V]
CONSUMPTION CURRENT [mA]

312

ID	RATED POWER [W]	RATED VOLTAGE [V]	RATED CURRENT [A]
0×1001	120	12.0	10.0
0×1002	160	16.0	10.0
...	...	...	...

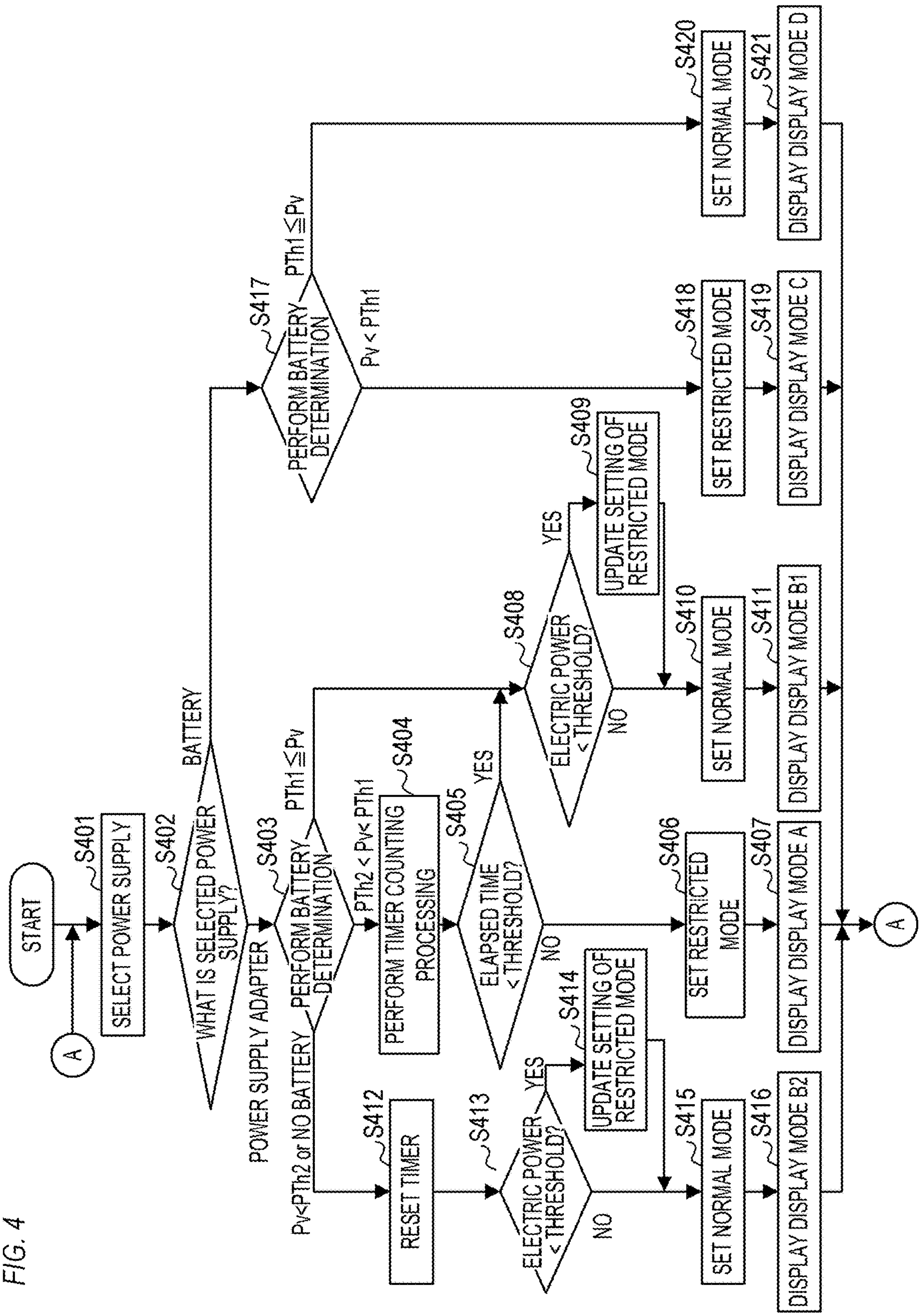

START
A
S401
SELECT POWER SUPPLY
S402
WHAT IS SELECTED POWER SUPPLY?
POWER SUPPLY ADAPTER
BATTERY
S403
PERFORM BATTERY DETERMINATION
Pv<PTh2 or NO BATTERY
PTh2 < Pv< PTh1
PTh1≦Pv
S404
PERFORM TIMER COUNTING PROCESSING
S405
ELAPSED TIME < THRESHOLD?
YES
NO
S406
SET RESTRICTED MODE
S407
DISPLAY DISPLAY MODE A
S408
ELECTRIC POWER < THRESHOLD?
YES
NO
S409
UPDATE SETTING OF RESTRICTED MODE
S410
SET NORMAL MODE
S411
DISPLAY DISPLAY MODE B1
S412
RESET TIMER
S413
ELECTRIC POWER < THRESHOLD?
YES
NO
S414
UPDATE SETTING OF RESTRICTED MODE
S415
SET NORMAL MODE
S416
DISPLAY DISPLAY MODE B2
S417
PERFORM BATTERY DETERMINATION
Pv < PTh1
PTh1≦Pv
S418
SET RESTRICTED MODE
S419
DISPLAY DISPLAY MODE C
S420
SET NORMAL MODE
S421
DISPLAY DISPLAY MODE D
A

FIG. 4

FIG. 5A

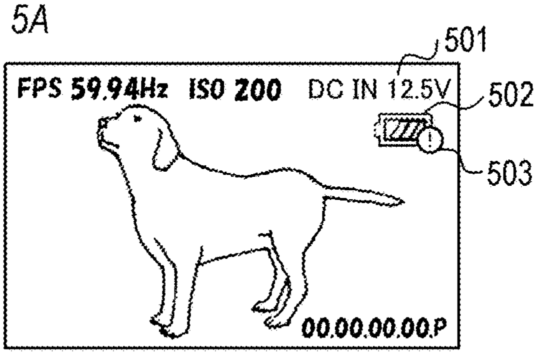


FIG. 5B

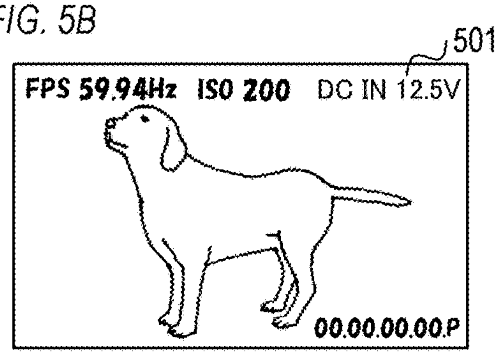


FIG. 5C

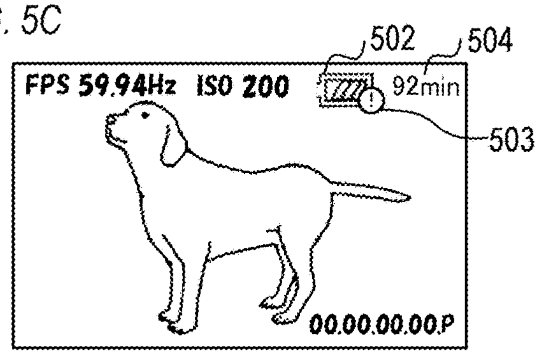


FIG. 5D

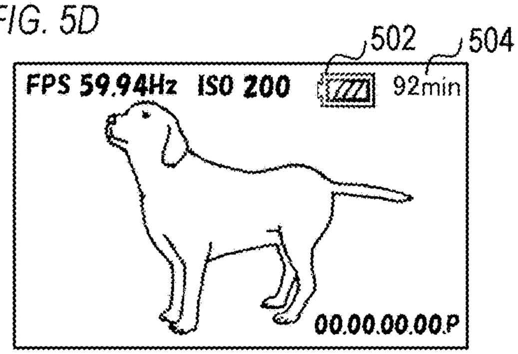
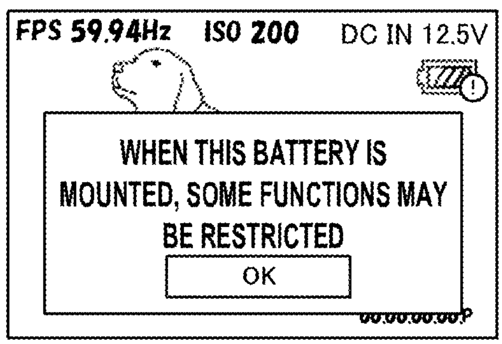


FIG. 6


FPS 59.94Hz
ISO 200
DC IN 12.5V
WHEN THIS BATTERY IS MOUNTED, SOME FUNCTIONS MAY BE RESTRICTED
OK

ELECTRONIC DEVICE, CONTROL METHOD OF ELECTRONIC DEVICE, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic device, a control method of an electronic device, and a non-transitory computer readable medium.

Description of the Related Art

[0002] As a power source of the imaging apparatus, there are a case where a chargeable/dischargeable battery (hereinafter, referred to as a battery) known as a lithium ion battery is connected, and a case where an adapter (hereinafter, referred to as a power supply adapter) that applies a DC voltage is connected to the imaging apparatus. A plurality of power supplies can be connected to the imaging apparatus in many cases. When a plurality of power supplies is connected to the imaging apparatus, it is desirable that imaging can be stably continued even if an electric power supplied from a certain power supply is reduced or impaired. JP 2020-154097 A discloses a method of determining a battery to be used for an operation of an imaging apparatus based on an operation mode of the imaging apparatus and a voltage of each battery when a plurality of batteries is connected to the imaging apparatus.

[0003] In addition, electric power that is suppliable to the imaging apparatus is different between the battery and the power supply adapter in many cases. JP 2006-115250 A discloses a method of determining an executable operation mode from among a plurality of operation modes based on whether electric power is supplied from a power supply adaptor when a plurality of power supplies are connected to an imaging apparatus.

[0004] However, in the technique disclosed in JP 2020-154097 A, when there is a battery in which the suppliable electric power is larger than the threshold voltage among the plurality of batteries connected to the imaging apparatus, the imaging apparatus receives supply of the electric power from the battery and operates in an operation mode requiring large power. However, the battery in use may be detached from the imaging apparatus due to a factor not intended by the user, and supply of the electric power from the battery may be instantaneously stopped. In the technique disclosed in JP 2006-115250 A, it is determined that the full operation mode can be executed if electric power is supplied from the power supply adaptor. However, the power supply adapter may be detached from the imaging apparatus due to a factor not intended by the user, and the supply of the electric power from the power supply adapter may be instantaneously stopped. In the techniques disclosed in JP 2020-154097 A and JP 2006-115250 A, when the supply of the electric power from the battery or the power supply adapter in use is instantaneously stopped due to a factor not intended by the user as described above, it is conceivable that another power supply connected to the imaging apparatus cannot cover the power necessary for the function in execution, and the function in execution cannot be normally stopped. For example, when supply of the electric power is instantaneously stopped due to an unintended factor during execution of the imaging function, it is conceivable that imaging is stopped. As a result, the recorded content is impaired.

SUMMARY OF THE INVENTION

[0005] The present invention provides a technique for preventing a function being executed from being stopped even when supply of the electric power is instantaneously stopped.

[0006] An electronic device according to the present invention includes: a processor; and a memory storing a program which, when executed by the processor, causes the electronic device to function as: an acquisition unit that acquires information on a power supply connected to the electronic device; a determination unit that determines a power supply to be used for an operation of the electronic device; and a setting unit that sets, as an operation mode of the electronic device, any of a plurality of operation modes including a first operation mode and a second operation mode in which a function that is usable is restricted as compared with the first operation mode, wherein in a case where a first power supply and a second power supply are connected to the electronic device, the acquisition unit acquires first information on the first power supply and second information on the second power supply, and the determination unit determines a power supply having a larger suppliable electric power among the first power supply and the second power supply as the power supply to be used for the operation of the electronic device based on the first information and the second information, and in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device, the setting unit sets the operation mode of the electronic device in accordance with a first electric power that is suppliable by the first power supply such that if the first electric power is larger than a threshold power, the first operation mode is set, and if the first electric power is smaller than the threshold power, the second operation mode is set.

[0007] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIGS. **1**A to **1**C are block diagrams illustrating a configuration of an imaging apparatus;

[0009] FIG. **2** is an explanatory diagram of electric power data;

[0010] FIG. **3** is an explanatory diagram of information on a power supply;

[0011] FIG. **4** is a flowchart illustrating an operation of an imaging apparatus;

[0012] FIGS. **5**A to **5**D are schematic diagrams of GUI screens; and

[0013] FIG. **6** is a schematic diagram of a warning screen.

DESCRIPTION OF THE EMBODIMENTS

[0014] In the description below, embodiments of the present invention are specifically described with reference to the accompanying drawings.

Configuration

[0015] FIG. **1**A is a block diagram illustrating a configuration of an imaging apparatus **100** according to the present embodiment. The imaging apparatus **100** is an example of an electronic device. The imaging apparatus **100** includes a control unit **101**, a memory **102**, a nonvolatile memory **103**, an image processing unit **104**, an imaging unit **105**, a lens unit **106**, a display unit **107**, a recording medium I/F **108**, an operation unit **109**, and a battery communication I/F **110**. The imaging apparatus **100** further includes a power supply selection unit **111**, a power supply unit **112**, an image output unit **114**, an accessory communication unit **115**, a network communication unit **116**, an audio input unit **117**, a battery connection unit **118**, and a connector **119**. The functional units included in the imaging apparatus **100** input and output data to and from each other via an internal bus **150**.

[0016] The control unit **101** is a central processing unit that controls each functional unit included in the imaging apparatus **100**. The control unit **101** includes at least one processor and/or at least one circuit. That is, the control unit **101** may be a processor, may be a circuit, or may be a combination of a processor and a circuit. For example, the control unit **101** controls each functional unit of the imaging apparatus **100** using the memory **102** as a work memory according to a program stored in the nonvolatile memory **103**.

[0017] The memory **102** is a volatile memory using a semiconductor element such as a random access memory (RAM).

[0018] The nonvolatile memory **103** stores image data, audio data, other data, and various programs for operating the control unit **101**, and the like. The nonvolatile memory **103** is configured with, for example, a hard disk (HD), a read only memory (ROM), and the like. The program as used herein includes programs for executing the flowcharts described below.

[0019] The image processing unit **104** performs image processing on image data stored in the nonvolatile memory **103** or a recording medium **131**, image data captured by the imaging unit **105**, or the like. Examples of the image processing that are executed by the image processing unit **104** include A/D conversion processing, D/A conversion processing, and encoding processing, compression processing, decoding processing, enlargement/reduction processing (resizing), noise reduction processing, and color conversion processing of image data. The image processing unit **104** may be configured with a dedicated circuit block for executing specific image processing. In addition, depending on the type of image processing, the control unit **101** can execute image processing according to a program without using the image processing unit **104**.

[0020] The imaging unit **105** is an imaging element such as a charge coupled device (CCD) sensor or a complementary MOS (CMOS) sensor. The imaging unit **105** can capture a still image or a moving image. The image data of the captured image is transmitted to the image processing unit **104**, subjected to various types of image processing, and then recorded in the nonvolatile memory **103** or the recording medium **131** as a still image file or a moving image file, respectively.

[0021] The lens unit **106** is a lens unit configured with a zoom lens, a focus lens, a shutter, an aperture, a distance measuring unit, an A/D converter, and the like. The lens unit **106** can adjust a zoom, a focus, a diaphragm, and the like.

[0022] The display unit **107** displays an image and a graphical user interface (GUI) screen configuring a GUI. A video signal or the like generated by each functional unit of the imaging apparatus **100** is input to the display unit **107**. The display unit **107** displays a video (image) based on the input video signal or the like. Note that, in the present embodiment, the imaging apparatus **100** includes the display unit **107**, but the display unit **107** may be a monitor, a projector, or the like outside the imaging apparatus **100**.

[0023] The image output unit **114** outputs an image, a GUI screen configuring a GUI, and the like to the outside. The image output unit **114** may output the same control signal as the display content of the display unit **107** to the outside or may output a control signal in which the display content is changed to the outside.

[0024] The recording medium I/F **108** is an interface with the recording medium **131** such as a memory card, a CD, or a DVD. The recording medium I/F **108** reads data from the recording medium **131** and writes data to the recording medium **131**.

[0025] For example, the operation unit **109** is an input device for receiving a user operation that includes a character information input device such as a keyboard, a pointing device such as a mouse or a touch panel, a button, a dial, a joystick, a touch sensor, and a touch pad. Note that the touch panel is an input device configured to overlap the display unit **107** in a planar manner and output coordinate information corresponding to a position being touched. The control unit **101** is notified of the input user operation.

[0026] The battery connection unit **118** has a connection mechanism for connecting a battery **120**. In addition, the battery connection unit **118** includes a plurality of terminals for connecting to a power supplying terminal and a communication terminal of the battery **120** when the battery **120** is connected. When the battery **120** is mounted to the battery connection unit **118**, an electric power from a battery cell **122** is supplied to the power supply selection unit **111** via the power supplying terminal. In addition, when the battery **120** is connected to the battery connection unit **118**, communication between a control unit **121** and the battery communication I/F **110** is performed via the communication terminal.

[0027] When the battery **120** is connected to the battery connection unit **118**, the battery communication I/F **110** communicates with the control unit **121** via the battery connection unit **118** and acquires information on the type of the battery **120**, the state of charge, the remaining amount, and the like of the battery cell **122**. In addition, the battery communication I/F **110** transmits various types of information and commands to the control unit **121**.

[0028] The power supply selection unit **111** is a power supply selection circuit that includes a power receiving unit receiving an electric power from the battery **120** and an electric power from a power supply adapter **123** and is configured with a comparator, a load switch, and the like. The power supply selection unit **111** selects (determines) one of the battery **120** and the power supply adapter **123** that can supply a higher electric power (power supply capability) as the power supply used for the operation of the imaging apparatus **100**. For example, when the power supply adapter **123** is connected to the connector **119**, the power supply selection unit **111** detects the output voltage of the power supply adapter **123**. In addition, the power supply selection unit **111** detects the output voltage of the battery **120** when the battery **120** is connected to the battery connection unit **118**. Then, when both the power supply adapter **123** and the battery **120** are connected, one of the power supply adapter

123 and the battery 120 having a higher output voltage is selected as the power source for the operation of the imaging apparatus 100. The control unit 101 performs control such that power from the selected one of the power supply adapter 123 and the battery 120 is used as the power source for the operation of the imaging apparatus 100 and power from the another of the power supply adapter 123 and the battery 120 is not used when the power is supplied from the selected one of the power supply adapter 123 and the battery 120. The control unit 101 performs control such that, in a case the power supply from the selected one of the power supply adapter 123 and the battery 120 which is being used as the power source for the operation of the imaging apparatus 100 is stopped, the power source for the operation of the imaging apparatus 100 is switched to the another of the power supply adapter 123 and the battery 120. The power supply selection unit 111 also acquires information on the connected battery 120 from the battery communication I/F 110 and transmits the information to the control unit 101. The control unit 101 performs control (display control) to display information on the connected battery 120.

[0029] The battery 120 is a power supply that supplies an electric power to the imaging apparatus 100. The battery 120 is a rechargeable battery (storage battery) and can be connected to (mounted to) and detached from the imaging apparatus 100. The battery 120 includes the control unit 121 and the battery cell 122. The amount of the electric power that the battery 120 can supply to the imaging apparatus 100 is limited by the amount of the electric power that the battery cell 122 has. In the battery 120, the electric power is supplied to the imaging apparatus 100 by the battery cell 122, and a charging state and a remaining capacity of the battery cell 122 are managed by the control unit 121. In the present embodiment, the battery 120 is connected to the battery connection unit 118 (and an internal terminal of the battery connection unit 118) provided inside the imaging apparatus 100. It can also be said that the battery 120 is an internal power supply of the imaging apparatus 100. Note that the battery 120 may be a non-rechargeable battery or may be a rechargeable battery provided integrally with the imaging apparatus 100.

[0030] The power supply adapter 123 is a power supply that supplies the electric power to the imaging apparatus 100. When the plug of the power supply adapter 123 is connected to the connector 119, the electric power from the power supply adapter 123 is supplied via the connector 119. Furthermore, the power supply adapter 123 can be connected to and detached from the imaging apparatus 100. The power supply adapter 123 converts, for example, the AC electric power supplied from a commercial power supply or the like into a DC electric power in a format suitable for the imaging apparatus 100 and supplies the converted power to the imaging apparatus 100. It can also be said that the power supply adapter 123 is connected to a power receiving unit (external terminal) provided outside the imaging apparatus 100, and the power supply adapter 123 that supplies the electric power from the outside of the imaging apparatus 100 is an external power source of the imaging apparatus 100.

[0031] The power supply unit 112 includes a DC-DC conversion circuit, converts a voltage from the power supply selected by the power supply selection unit 111 into a voltage necessary for each block of the imaging apparatus 100, and supplies the voltage to each block for a necessary period. The power supplying processing of the power supply unit 112 is controlled by the control unit 101 as described below.

[0032] The accessory communication unit 115 is a communication unit for communication with an accessory device mounted on the imaging apparatus 100. The accessory communication unit 115 acquires information from the accessory device or controls the accessory device according to an instruction of the control unit 101. For example, the accessory communication unit 115 causes the LED light mounted on the imaging apparatus 100 to emit light.

[0033] The network communication unit 116 includes a wireless communication circuit, a local area network (LAN) communication terminal, and the like and connects the imaging apparatus 100 to a network to perform various communications.

[0034] The audio input unit 117 receives an audio from an internal microphone provided in the imaging apparatus 100 or an external microphone connected to an audio input terminal. The audio signal input by the audio input unit 117 is processed as audio data by the control unit 101.

[0035] FIG. 1B is an example of a configuration of a main part of the imaging apparatus 100 in a case where a USB terminal 130 and a PD control unit 113 are used instead of the connector 119 in FIG. 1A.

[0036] The USB terminal 130 is a terminal conforming to the Universal Serial Bus (USB) standard and can receive the electric power supplied from an external USB power supply. Here, the USB terminal 130 is a USB Type-C terminal. An external power supplying device conforming to the USB Power Delivery (USB PD) standard is connected to the USB terminal 130.

[0037] When the power supplying device is connected to the USB terminal 130, the PD control unit 113 performs power receiving processing conforming to the USB PD standard. The PD control unit 113 performs predetermined negotiation (power supplying negotiation) with a power supplying device connected to the imaging apparatus 100 through communication conforming to the USB PD standard. The PD control unit 113 receives information on the electric power including voltage and current that is suppliable by the power supplying device connected to the USB terminal 130 and transmits the information on the electric power including voltage and current that is suppliable by the power supplying device connected to the power supply selection unit 111.

[0038] When both the battery 120 and the power supplying device are connected, the power supply selection unit 111 may determine the power supply having the larger voltage among the voltage that is suppliable by the power supplying device connected to the USB terminal 130 and the output voltage of the battery 120 as the power supply used for the operation of the imaging apparatus 100.

[0039] FIG. 1C is an example of a configuration of a main part of the imaging apparatus 100 in a case where a battery coupler 124 and an AC adapter 127 are used instead of the connector 119 and the power supply adapter 123 of FIG. 1A.

[0040] The battery coupler (DC coupler) 124 includes a control unit 125 and a DC conversion unit 126. The battery coupler 124 has the same shape as the battery 120. The battery 120 and the battery coupler 124 are connected to the imaging apparatus 100 in a state of being mounted to a battery grip (not illustrated). When the battery grip including the battery 120 and the battery coupler 124 is mounted to the imaging apparatus **100**, the battery communication I/F **110** communicates with the control unit **121** and the control unit **125** to acquire various types of information from the battery **120** and the battery coupler **124**. The control unit **101** can recognize the type of the battery coupler **124** based on the information from the battery communication I/F **110**. The DC conversion unit **126** converts the electric power supplied from the AC adapter **127** into an electric power suitable for the imaging apparatus **100** and supplies the power to the imaging apparatus **100**. The AC adapter **127** is a power supply that supplies the electric power to the battery coupler **124**. When the plug of the AC adapter **127** is connected to a connector (not illustrated) of the battery coupler **124**, the electric power from the AC adapter **127** is supplied to the DC conversion unit **126**. The AC adapter **127** can be connected to and detached from the battery coupler **124**. The AC adapter **127** supplies, for example, AC power supplied from a commercial power supply or the like to the battery coupler **124**.

[0041] The power supply selection unit **111** may detect the output voltages of the battery **120** and the battery coupler **124** and determine the power supply having the larger voltage that is suppliable, among the battery **120** and the battery coupler **124**, as the power supply used for the operation of the imaging apparatus **100**.

Data on Electric Power

[0042] The imaging apparatus **100** has at least two operation (imaging) modes. One is a mode (Operation Mode **1**, normal mode) in which all functions of the imaging apparatus **100** can be used. The other is a mode (a mode after Operation Mode **2**, restricted mode) in which usable functions are limited as compared with the normal mode. In the restricted mode, control is performed so that some predetermined functions among functions usable in the normal mode cannot be used (use is prohibited).

[0043] FIG. **2** is an explanatory diagram of power data **201** and **202** which is data on an electric power. The control unit **101** acquires information on power consumption in each operation mode based on the power data **201** and the power data **202** recorded in the nonvolatile memory **103**.

[0044] The power data **201** includes information on power consumption in a case where each function of the imaging apparatus **100** is used (turned ON) and in a case where each function is not used (turned OFF). Each function is, for example, a network, accessory shoe power supply, external microphone input, and lens power supply. The electric power is supplied to the network communication unit **116** when the network is ON, to the accessory communication unit **115** when accessory shoe power supply is ON, to the audio input unit **117** when external microphone input is ON, and to the lens unit **106** when lens power supply is ON. The function and power consumption illustrated in the power data **201** are merely examples. For example, the power data **201** may include power consumption when functions such as high luminance display of the display unit **107** and output by the image output unit **114** are turned ON and OFF.

[0045] The power data **202** includes information on total electric power (sum of power consumption of each function) in each operation mode. In the power data **202**, the functions of the network, the accessory shoe power supply, the external microphone input, and the lens power supply are shown as ON/OFF, but this is also merely an example. In the power data **202**, information may be recorded not in binary such as ON/OFF but in a continuous data string, a function format, or the like.

[0046] In Operation Mode **1**, all functions of the imaging apparatus **100** can be used. A total electric power **P1** in Operation Mode **1** in which all the functions are turned on is 68 [W] (in the power data **201**, 5+3+2+8+50=68). In Operation Mode **2**, the functions of the accessory shoe power supply and the external microphone input are turned OFF, and other functions are turned ON. A total electric power **P2** in Operation Mode **2** is 63 [W]. In Operation Mode **3**, functions of the accessory shoe power supply and the lens power supply are turned OFF, and other functions are turned ON. A total electric power **P3** in Operation Mode **3** is 57.3 [W]. As described above, in Operation Mode **2**, since the functions that can be used in Operation Mode **1** are restricted, the power consumption is smaller than that in Operation Mode **1** by the electric power necessary for the restricted functions. Also, in Operation Mode **3**, since the functions that can be used in Operation Mode **1** are restricted, the power consumption is smaller than that in Operation Mode **1** by the electric power necessary for the restricted functions.

Data on Power Supply

[0047] FIG. **3** is an explanatory diagram of data on a power supply. Battery information **301** is information acquired by the power supply selection unit **111** (control unit **101**) from the battery **120** connected to the imaging apparatus **100**. The battery information **301** includes a battery ID, a capacity [%], a voltage [V], a consumption current [mA], and a dischargeable remaining amount [mAh]. The power supply selection unit **111** calculates the electric power that is suppliable by the battery **120**, for example, from the voltage [V] and the consumption current [mA].

[0048] Further, the power supply selection unit **111** may acquire the electric power that is suppliable by the battery **120** with reference to information table **302**. The information table **302** indicates a rated power [W], a rated voltage [V], and a rated current [A] for each battery ID and is recorded in the nonvolatile memory **103**. The power supply selection unit **111** refers to the information table **302** based on the battery ID acquired from the battery information **301** and can acquire an electric power that is suppliable by the battery **120**.

[0049] The power supply selection unit **111** can acquire the remaining amount of the battery **120** from the capacity [%] of the battery information **301**. Furthermore, the power supply selection unit **111** can calculate the drivable time of the imaging apparatus **100** by the following calculation formula (Formula 1).

$$\text{Drivable time (min)} = \text{Dischargeable remaining amount (mAh)}/\text{Current consumption (mA)} \times 60 \quad \text{(Equation 1)}$$

[0050] Coupler information **311** is information acquired by the power supply selection unit **111** from the battery coupler **124** connected to the imaging apparatus **100**. The coupler information **311** includes a battery coupler ID, a voltage [V], and a consumption current [mA]. The information table **312** indicates a rated power [W], a rated voltage

[V], and a rated current [A] for each battery coupler ID and is recorded in the nonvolatile memory **103**. The power supply selection unit **111** may calculate the electric power that is suppliable by the battery coupler **124** from the voltage [V] and the consumption current [mA] in the coupler information **311** or may acquire the electric power by referring to the information table **312**.

[0051] FIG. **4** is a flowchart illustrating an example of the operation of the imaging apparatus **100**. The operation is implemented by loading a program recorded in the nonvolatile memory **103** into the memory **102** and executing the program by the control unit **101** For example, when the power supply of the imaging apparatus **100** is turned on, the operation of FIG. **4** is started and repeatedly executed. The repetition timing may be controlled by synchronous processing by a periodic timer or may be controlled by asynchronous processing triggered by an event related to a series of operations. Furthermore, in the following description, as illustrated in FIG. **1**A, the imaging apparatus **100** to which the battery **120** and the power supply adapter **123** can be connected is described. In addition, Operation Mode **1** illustrated in FIG. **2** is set as the normal mode, and Operation Mode **2** and Operation Mode **3** are set as the restricted modes.

[0052] In step S**401**, the control unit **101** (power supply selection unit **111**) determines the power supply used for the operation of the imaging apparatus **100**. When the battery **120** and the power supply adapter **123** are connected, as described above, the power supply selection unit **111** detects the output voltage of the battery **120** and the output voltage of the power supply adapter **123**, and sends the detection result to the control unit **101**. The control unit **101** determines a power supply having a larger output voltage (a power supply having a larger suppliable electric power) among the battery **120** and the power supply adapter **123** as a power supply used for the operation of the imaging apparatus **100**. When the battery **120** is connected and the power supply adapter **123** is not connected, the battery **120** is selected as the power supply. In addition, when the power supply adapter **123** is connected and the battery **120** is not connected, the power supply adapter **123** is selected as the power supply.

[0053] Note that as illustrated in FIG. **1**B, when the battery **120** is connected and the power supplying device is connected to the USB terminal **130**, the control unit **101** communicates with the battery **120** and the power supplying device connected to the USB terminal **130** to acquire information on the power supply capability. In addition, as illustrated in FIG. **1**C, when the battery **120** and the battery coupler **124** are connected, information on the power supply capability is acquired by performing communication with the battery **120** and the battery coupler **124**. The information on the battery **120** may be information indicating electric power that is suppliable by the battery **120** or may be information indicating an ID (identifier) of the battery **120**. When the information indicating the ID of the battery **120** is acquired, the control unit **101** refers to the information table **302** (**312**) in FIG. **3** and acquires an electric power that is suppliable by the battery **120**.

[0054] In step S**402**, the control unit **101** determines whether the power supply determined in step S**401** is the battery **120**. The processing proceeds to step S**417** when the power supply determined in step S**401** is the battery **120**, and the processing proceeds to step S**403** when the power source is the power supply adapter **123**. Note that the determination in step S**402** may be interpreted as determination of whether the power supply used for the operation of the imaging apparatus **100** is a battery.

[0055] In step S**403**, the control unit **101** compares an electric power (Pv) that is suppliable from the battery **120** with threshold powers PTh**1** and PTh**2**. Here, the rated power illustrated in FIG. **3** is an electric power that is suppliable by the battery **120**. The processing proceeds to step S**408** when the electric power that is suppliable by the battery **120** is the threshold power PTh**1** or more, and the processing proceeds to step S**404** when the electric power is larger than the threshold power PTh**2** and smaller than the threshold power PTh**1**. When the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**2**, or when the battery **120** is not connected and only a power supply (for example, the power supply adapter **123**) other than the battery **120** is connected, the processing proceeds to step S**412**.

[0056] The threshold power PTh**1** is a value larger than the total electric power P**1** (**68** [W]) in the normal mode (Operation Mode **1** in FIG. **2**), and is, for example, 70 [W]. The threshold power PTh**2** is a lower limit value of the power consumption necessary for operating in the restricted mode (Operation Mode **2** or Operation Mode **3** in FIG. **2**). When there are a plurality of restricted modes, the threshold power PTh**2** may be a lower limit value of the total electric power of a specific restricted mode or may be a lower limit value of the total electric power of a mode having the highest total electric power among the restricted modes (for example, a value larger than P**2** and smaller than P**1**). The threshold powers PTh**1** and PTh**2** may be determined in consideration of the stability and error of the power supply.

[0057] In step S**404**, the control unit **101** performs processing (timer counting processing) of measuring elapsed time from attachment of the battery **120**. The timer count processing may be interpreted as processing of measuring elapsed time after it is determined that the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**1**. When the processing of step S**404** is executed for the first time after step S**412**, the control unit **101** records the current time acquired from the system clock as the power supply attachment time in the memory **102**. When executing the processing of step S**404** again, the control unit **101** calculates the difference between the updated current time and the power supply attachment time and measures the elapsed time.

[0058] In step S**405**, the control unit **101** determines whether the elapsed time measured in step S**404** is less than the threshold time. When the elapsed time is less than the threshold time, the processing proceeds to step S**408**. Otherwise, the processing proceeds to step S**406**. The threshold time is, for example, 4 seconds. The threshold time may be determined based on the time required for attaching and detaching the battery **120**.

[0059] In step S**406**, the control unit **101** sets the restricted mode. When there are a plurality of restricted modes, a specific restricted mode may be set, or a restricted mode selected by the user may be set. In addition, a mode of the setting updated by the processing in step S**409** or step S**414** described below may be set.

[0060] In step S**407**, the control unit **101** displays the GUI screen in a display mode A. The display mode A is described below with reference to FIG. **5**A.

[0061] In step S**408**, the control unit **101** detects the power consumption in the current operation mode of the imaging apparatus **100**. Here, the control unit **101** sets the operation mode to the normal mode until the operation mode is set in any one of S**415**, S**406**, S**410**, S**418**, and S**420** after the power is turned on. Then, the control unit **101** determines whether the power consumption in the current operation mode is smaller than a threshold power PTh**3**. The processing proceeds to step S**409** when the current power consumption is smaller than the threshold power PTh**3**. Otherwise, the processing proceeds to step S**410**. The threshold power PTh**3** used for the determination in step S**408** is, for example, the threshold power PTh**1**, but may be a value determined in consideration of a calculation error of power consumption. The threshold power PTh**3** is, for example, a value of the threshold power PTh**1** or less.

[0062] In step S**409**, the control unit **101** updates the setting of the restricted mode so that the function executed in the state where the power consumption is smaller than the threshold power PTh**3** in the normal mode can be executed. For example, the control unit **101** updates the setting of the restricted mode so that the function currently set to be turned ON is set to be turned ON in preference to other functions. As a result, when the setting is switched from the normal mode to the restricted mode, it is possible to reduce the time and effort required for the user to set the function executed in the normal mode to be executable even in the restricted mode.

[0063] In step S**410**, the control unit **101** sets the normal mode.

[0064] In step S**411**, the control unit **101** displays the GUI screen in a display mode B**1**. The display mode B**1** is described below with reference to FIG. **5**B.

[0065] According to the processing of steps S**401** to S**403**, S**406**, and S**410**, control is performed as follows. When the power supply adapter **123** is determined as the power supply to be used for the operation of the imaging apparatus **100**, and the battery **120** is connected, if the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**, the control unit **101** sets the normal mode. When the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**1**, the control unit **101** sets the restricted mode. The control unit **101** sets the operation mode of the imaging apparatus **100** based on the electric power of the power supply (the power supply having the smaller suppliable electric power) that is not the power supply used for the operation of the imaging apparatus **100**. As a result, even when the plug of the power supply adapter **123** is unplugged from the connector **119**, the supply of the electric power from the power supply adapter **123** is momentarily interrupted, and the power supply used in the imaging apparatus **100** is switched to the battery **120** which has the suppliable power smaller than the threshold power PTh**1**, the electric power supply for maintaining the operation of the imaging apparatus **100** in the restricted mode can be covered by the battery **120**, and the function being executed by the imaging apparatus **100** can be prevented from being stopped.

[0066] According to the processing of steps S**404** to S**406** and S**410**, the control unit **101** sets the normal mode until the elapsed time from the determination that the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**1** reaches the threshold time. The control unit **101** sets the restricted mode when the elapsed time reaches the threshold time. Since the switching to the restricted mode is suspended until the elapsed time reaches the threshold time, the function can be prevented from being restricted when a battery having a suppliable electric power smaller than the threshold power PTh**1** is erroneously mounted to the imaging apparatus **100**.

[0067] In step S**412**, the control unit **101** resets the power supply attachment time recorded in step S**404**.

[0068] The processing in steps S**413** to S**415** is the same as the processing in steps S**408** to S**410**. The control unit **101** sets the normal mode when only the power supply adapter **123** is connected to the imaging apparatus **100**.

[0069] In step S**416**, the control unit **101** displays the GUI screen in a display mode B**2**. The display mode B**2** is described below with reference to FIG. **5**B.

[0070] In step S**417**, the control unit **101** compares the electric power (Pv) that is suppliable from the battery **120** with threshold power PTh**1**. When the electric power that is suppliable by the battery **120** is the threshold power PTh**1** or more, the processing proceeds to step S**420**. Otherwise, the processing proceeds to step S**418**. Although not shown in S**417**, when the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**2**, it is desirable to stop this processing flow.

[0071] The processing in steps S**418** and S**420** is the same as the processing in steps S**406** and S**409**. When the battery **120** is determined as the power supply to be used for the operation of the imaging apparatus **100**, if the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**, the control unit **101** sets the normal mode. When the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**1**, the control unit **101** sets the restricted mode.

[0072] In step S**419**, the control unit **101** displays the GUI screen in a display mode C. The display mode C is described below with reference to FIG. **5**C.

[0073] In step S**421**, the control unit **101** displays the GUI screen in a display mode D. The display mode D is described below with reference to FIG. **5**D.

[0074] When the power supply adapter **123** is selected as the power supply to be used by the imaging apparatus **100**, the power supply selection unit **111** determines whether the electric power is supplied from the power supply adapter **123**. Also, when determining that the supply of the electric power from the power supply adapter **123** is stopped, the power supply selection unit **111** notifies the control unit **101** of the stop, selects the battery **120** as a power supply, and sends the electric power from the battery **120** to the power supply unit **112**. As described above, when the power supply adapter **123** is selected as a power supply, if the power supply adapter **123** is removed, the power supply is switched to the battery **120** and processing of the imaging apparatus **100** is continued using the power from the battery **120**.

[0075] In the example of FIG. **4**, as illustrated in FIG. **1**A, the imaging apparatus that can connect the power supply adapter **123** and the battery **120** is described, but the present invention is not limited thereto, and the configurations of FIGS. **1**B and **1**C can also be adopted.

[0076] When the configuration of FIG. **1**B is adopted, in S**401**, the control unit **101** determines which power supply is to be used as the power supply of the imaging apparatus **100** based on the electric power of the battery **120** and the information of the electric power that is suppliable by the power supplying device connected to the USB terminal **130**.

Then, when the selected power supply is the power supplying device connected to the USB terminal **130**, the processing of S**403** and subsequent processing are performed. In this case, the power supply adapter **123** in FIG. **4** is processed as a power supplying device connected to the USB terminal **130**.

[0077] Furthermore, when the configuration of FIG. **1**C is adopted, in S**401**, the control unit **101** determines which power supply is to be used as the power supply of the imaging apparatus **100** based on the information of the electric power that is suppliable by the battery **120** and the information of the electric power that is suppliable by the battery coupler **124**. Also, when the selected power supply is the battery coupler **124**, the processing of S**403** and subsequent processing are performed. In this case, the processing is performed using the power supply adapter **123** of FIG. **4** as the battery coupler **124**.

[0078] Furthermore, when the configuration of FIGS. **1**B and **1**C is adopted, in step S**401**, the control unit **101** determines the power supply used for the operation of the imaging apparatus **100** based on the information of the electric power that is suppliable by the power supply connected to the imaging apparatus **100**, but may determine the power supply used for the operation of the imaging apparatus **100** based on the identification information of the power supply. For example, when the configuration of FIG. **1**C is adopted, the control unit **101** may determine the battery coupler **124** as the power supply used for the operation of the imaging apparatus **100** in preference to the battery **120** based on the identification information.

[0079] Furthermore, in any configuration of FIGS. **1**A to **1**C, the control unit **101** may determine a predetermined power supply among a plurality of connected power supplies as the power supply used for the operation of the imaging apparatus **100**.

[0080] For example, in the configuration of FIG. **1**A, when the battery **120** and the power supply adapter **123** are connected, the power supply adapter **123** may be determined as the power supply used for the operation of the imaging apparatus **100**. Note that, when the power supply adapter **123** is selected as the power supply used for the operation of the imaging apparatus **100**, if the electric power that is suppliable by the power supply is smaller than the threshold power PTh**2**, the control unit **101** may set the restricted mode instead of the normal mode in step S**415**.

[0081] Further, the configuration of FIG. **1**A may be a configuration in which the plurality of batteries **120** can be connected. For example, when the power supply adapter **123** is not connected, and two batteries are connected as the battery **120**, in step S**403**, the control unit **101** compares the electric power suppliable by the battery having the smaller suppliable electric power with the threshold powers PTh**1** and PTh**2**. This is to reduce the risk of instantaneous power interruption of the imaging apparatus **100** by switching the power supply of the imaging apparatus **100** to the other battery when the supply of the electric power from the battery is stopped while the imaging apparatus **100** operates using the battery having the larger suppliable electric power. In addition, when the power supply adapter **123** is connected, and two batteries are connected as the battery **120**, in step S**403**, the control unit **101** compares the power that is suppliable by the battery having the larger electric power that is suppliable, with the threshold powers PTh**1** and PTh**2**. This is because the imaging apparatus **100** operates using the power supply adapter **123**, but when the supply of the electric power from the power supply adapter **123** is stopped, the imaging apparatus **100** operates using the battery having the larger power that is suppliable.

[0082] FIGS. **5**A to **5**D are schematic diagrams of a GUI screen displayed on the display unit **107**.

[0083] FIG. **5**A is a schematic diagram of a GUI screen in the display mode A. The display mode A is a mode that is displayed, for example, when the power supply adapter **123** is determined as the power to be used for the operation of the imaging apparatus **100**, and the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**2** and smaller than the threshold power PTh**1**. In an upper portion of the display unit **107**, adapter information **501** "DC IN 12.5 V" is displayed as information on the power supply adapter **123**.

[0084] In addition, under the adapter information **501**, a battery icon **502** is displayed as information on the battery **120**. Since the electric power that is suppliable from the battery **120** is smaller than the threshold power PTh**1**, as the information on the battery **120**, a mark **503** indicating "!" is also displayed. In the example of FIG. **5**A, the mark **503** is displayed on the battery icon **502** in a superimposed manner. However, a mark other than "!" may be displayed in a superimposed manner. A battery icon in a form (for example, different shapes or colors) different from the battery icon **502** may be displayed. As a result, it is possible to indicate to the user that the mounted battery is different from a normal battery (a battery having a suppliable electric power larger than the threshold power PTh**1**).

[0085] FIG. **5**B is a schematic diagram of a GUI screen in the display mode B**1**. The display mode B**1** is a mode that is displayed, for example, when the power supply adapter **123** is determined as the power to be used for the operation of the imaging apparatus **100**, and the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**. The adapter information **501** is displayed on the upper portion of the display unit **107**. Since the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**, the information on the battery **120** is not displayed in the display mode B**1**.

[0086] FIG. **5**B is also a schematic diagram of a GUI screen in the display mode B**2**. The display mode B**2** is a mode that is displayed, for example, when the power supply adapter **123** is determined as a power supply used for the operation of the imaging apparatus **100**, and the battery **120** is not mounted (alternatively, the electric power that is suppliable by the battery **120** is smaller than the threshold power PTh**2**).

[0087] Although the GUI screen in the display mode B**1** and the GUI screen in the display mode B**2** are the same, the GUI screens may not be the same. For example, an item indicating that a normal battery is mounted may be displayed in the display mode B**1**, and the item may not be displayed in the display mode B**2**.

[0088] FIG. **5**C is a schematic diagram of a GUI screen in the display mode C. The display mode C is a mode that is displayed, for example, when the battery **120** is determined as the power supply to be used for the operation of the imaging apparatus **100**, and the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**2** and smaller than the threshold power PTh**1**.

[0089] When the battery **120** is determined as the power supply used for the operation of the imaging apparatus **100**

(display mode C) and when the power supply adapter **123** is determined as the power supply (display mode A), different information is displayed as the information on the battery **120**. In FIG. **5**C, as the information on the battery **120**, time **504** that is a period of time during which the battery **120** can be driven is displayed in addition to the battery icon **502** and the mark **503**. A battery icon in a form different from the battery icon **502** may be displayed. As a result, it is possible to indicate to the user that the battery **120** is a power supply used for the operation of the imaging apparatus **100**.

[0090] FIG. **5**D is a schematic diagram of a GUI screen in the display mode D. The display mode D is a mode that is displayed, for example, when the battery **120** is determined as the power supply to be used for the operation of the imaging apparatus **100**, and the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**. In the display mode D, the battery icon **502** and the time **504** are displayed, but the mark **503** is not displayed because the electric power that is suppliable by the battery **120** is larger than the threshold power PTh**1**.

[0091] Note that in each display mode, when there is a function set to be turned OFF, the function may be displayed in a form different from the case where the function is set to be turned ON. For example, the icon indicating the function to be set to be turned OFF may be displayed in red. As a result, it is possible to indicate the presence or absence of the restriction of the function to the user.

[0092] FIG. **6** is a schematic diagram of a warning screen displayed on the display unit **107**. The warning screen displays information on the restricted mode. In the example of FIG. **6**, a text indicating that some functions are limited is displayed on the warning screen. The warning screen is displayed for a predetermined period of time at a timing when the restricted mode is first set after the battery **120** having the suppliable electric power larger than the threshold power PTh**2** and smaller than the threshold power PTh**1** is mounted on the imaging apparatus **100**, and the imaging apparatus **100** is activated, and then the display is stopped. Note that, after being displayed once at this timing, the warning screen may not be displayed again until the imaging apparatus **100** is initialized. In addition, the warning screen may include other information, for example, information such as a set operation mode, a list of restricted functions, and threshold time determined in step S**405**. The battery icon **502** may be controlled to blink at the timing when the warning screen is displayed. By displaying the warning screen in addition to the GUI screen illustrated in FIGS. **5**A to **5**D, it is possible to more clearly indicate the set operation mode to the user.

[0093] As described above, in the present embodiment, when a plurality of power supplies are connected to the imaging apparatus **100**, the operation mode of the imaging apparatus **100** is set based on the electric power that is suppliable by the power supply having the smaller power that is suppliable. As a result, the risk of instantaneous power interruption of the imaging apparatus **100** can be reduced. In addition, information on the power supply is displayed according to each operation mode. As a result, it is possible to indicate the set operation mode to the user.

[0094] Note that the above-described various types of control may be processing that is carried out by one piece of hardware (e.g., processor or circuit), or otherwise. Processing may be shared among a plurality of pieces of hardware (e.g., a plurality of processors, a plurality of circuits, or a combination of one or more processors and one or more circuits), thereby carrying out the control of the entire device.

[0095] Also, the above processor is a processor in the broad sense, and includes general-purpose processors and dedicated processors. Examples of general-purpose processors include a central processing unit (CPU), a micro processing unit (MPU), a digital signal processor (DSP), and so forth. Examples of dedicated processors include a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a programmable logic device (PLD), and so forth. Examples of PLDs include a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and so forth.

[0096] The embodiment described above (including variation examples) is merely an example. Any configurations obtained by suitably modifying or changing some configurations of the embodiment within the scope of the subject matter of the present invention are also included in the present invention. The present invention also includes other configurations obtained by suitably combining various features of the embodiment.

[0097] Furthermore, in the above-described embodiment, the case where the present invention is applied to the imaging apparatus is described as an example, but the present invention is not limited to this example and can be applied to any electronic device as long as a plurality of power supplies can be connected to the electronic device. For example, the present invention can also be applied to a portable game machine.

[0098] According to the present invention, even when the supply of the electric power is instantaneously stopped, the function being executed is not stopped.

OTHER EMBODIMENTS

[0099] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0100] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0101] This application claims the benefit of Japanese Patent Application No. 2024-031911, filed on Mar. 4, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An electronic device comprising:
a processor; and
a memory storing a program which, when executed by the processor, causes the electronic device to function as:
an acquisition unit that acquires information on a power supply connected to the electronic device;
a determination unit that determines a power supply to be used for an operation of the electronic device; and
a setting unit that sets, as an operation mode of the electronic device, any of a plurality of operation modes including a first operation mode and a second operation mode in which a function that is usable is restricted as compared with the first operation mode, wherein
in a case where a first power supply and a second power supply are connected to the electronic device,
the acquisition unit acquires first information on the first power supply and second information on the second power supply, and
the determination unit determines a power supply having a larger suppliable electric power among the first power supply and the second power supply as the power supply to be used for the operation of the electronic device based on the first information and the second information, and
in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device, the setting unit sets the operation mode of the electronic device in accordance with a first electric power that is suppliable by the first power supply such that
if the first electric power is larger than a threshold power, the first operation mode is set, and
if the first electric power is smaller than the threshold power, the second operation mode is set.

2. The electronic device according to claim **1**, wherein
the information on the power supply connected to the electronic device is an output voltage of the power supply, and the determination unit determines a power supply having a higher output voltage among the first power supply and the second power supply as the power supply to be used for the operation of the electronic device.

3. The electronic device according to claim **1**, wherein
the information on the power supply connected to the electronic device indicates electric power that is suppliable by the power supply.

4. The electronic device according to claim **1**, wherein
the information on the power supply connected to the electronic device indicates an identifier of the power supply.

5. The electronic device according to claim **1**, wherein
the first power supply is a battery, the second power supply is a power supply adapter that supplies electric power from a commercial power supply, and
the first electric power is a rated power of the battery.

6. The electronic device according to claim **1**, wherein
in a case where the second power supply is connected to the electronic device, and the first power supply is not connected to the electronic device, the first operation mode is set.

7. The electronic device according to claim **1**, wherein
the program, when executed by the processor, further causes the electronic device to function as a display control unit that performs control to display information on the first power supply.

8. The electronic device according to claim **7**, wherein
in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device,
if the first electric power is larger than the threshold power, the display control unit performs control so that the information on the first power supply is not displayed, and
if the first electric power is smaller than the threshold power, the display control unit performs control so that the information on the first power supply is displayed.

9. The electronic device according to claim **8**, wherein
the display control unit performs control so that information to be displayed as the information on the first power supply differ between a case where the first power supply is determined as the power supply to be used for the operation of the electronic device and a case where the second power supply is determined as the power supply to be used for the operation of the electronic device.

10. The electronic device according to claim **1**, wherein
the electronic device is an imaging apparatus.

11. A control method of an electronic device comprising:
acquiring information on a power supply connected to the electronic device;
determining a power supply to be used for an operation of the electronic device; and
setting, as an operation mode of the electronic device, any of a plurality of operation modes including a first operation mode and a second operation mode in which a function that is usable is restricted as compared with the first operation mode, wherein
in a case where a first power supply and a second power supply are connected to the electronic device,
first information on the first power supply and second information on the second power supply are acquired, and
among the first power supply and the second power supply, a power supply having a larger suppliable electric power is determined as the power supply to be used for the operation of the electronic device based on the first information and the second information, and
in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device,
if a first electric power that is suppliable by the first power supply is larger than a threshold power, the first operation mode is set, and

if the first electric power is smaller than the threshold power, the second operation mode is set.

12. A non-transitory computer readable medium that stores a program, wherein the program causes a computer to execute a control method of an electronic device, the control method comprising:

acquiring information on a power supply connected to the electronic device;

determining a power supply to be used for an operation of the electronic device; and

setting, as an operation mode of the electronic device, any of a plurality of operation modes including a first operation mode and a second operation mode in which a function that is usable is restricted as compared with the first operation mode, wherein

in a case where a first power supply and a second power supply are connected to the electronic device,

first information on the first power supply and second information on the second power supply are acquired, and

among the first power supply and the second power supply, a power supply having a larger suppliable electric power is determined as the power supply to be used for the operation of the electronic device based on the first information and the second information, and

in a case where the second power supply is determined as the power supply to be used for the operation of the electronic device,

if a first electric power that is suppliable by the first power supply is larger than a threshold power, the first operation mode is set, and

if the first electric power is smaller than the threshold power, the second operation mode is set.

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